

Loanword Adaptation Algorithm for Filipino

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Abstract—This is the abstract.
Index Terms—one, two, three

I. INTRODUCTION

Borrowing words from one language (or the source language L_2) to another (the borrowing language L_1) is a common linguistic phenomenon. For English, the word *modus operandi* may be used in criminal investigations. In the context of shipping, one may use *cruise*, *dock*, or *yacht*, all of which are from Dutch.

Borrowing is more common in locales where multiple languages are being used simultaneously in similar contexts. An example of this is the Philippine context: which has both Filipino ($\mathcal{F}$) and English ($\mathcal{E}$) as their official language. In casual contexts, like going to school, one may use *paaralan* which is the native Filipino word for “school”. Or borrow by using *eskuwelahan* from Spanish, or *iskul* or *skul* from English. Furthermore, the Philippines also has a rich history of borrowings from Malay and Chinese way before they started borrowing from Spanish and English (during their colonial period) [1].

In general, L_1 borrows a word $w \in L_2$, we consider how the w is pronounced and written in L_2 and constraints in L_1 [2], [3]. Using the example above, we look at the base pronunciation and spelling of *school*, which are /sku:l/ and “school”. With the relevant Filipino constraints: “i/e-” prefix for English or Spanish loanwords and strict phoneme-grapheme mapping. Which is why the adapted loanword is *iskul*.

The algorithmic task of transforming $w \in L_2$ to its adapted form $\omega \in \mathcal{F}$ has importance in Filipino-centric NLP where loanword adaptation or code-switching occurs, like spell and grammar checking for Filipino text data with loanwords which is common in a most contexts such as social media posts, casual conversations, and technical settings where using the native words may be difficult. Or, text-to-speech or sentence generation for Taglish.

Finally, even though Filipino has a orthography manual publicly available, no software solution is yet to be implemented for writing adapted loanwords.

Hence, the goal of the project is to study loanword adaptation (LA) in Filipino and implementing an adaptation algorithm.

A. Phonology of Filipino

B. Loanwords in Filipino

II. LOANWORD ADAPTATION AS A TRANSDUCER

We can think of the computational task of LA as a string transduction problem

$$L_2 \mapsto F^*$$

where F is the alphabet of $\mathcal{F}$ which is the same as the latin alphabet but with the letter Ñ and Ng added. Note that ng, as the velar nasal ŋ, is counted as a single character.

By framing it as a transduction problem, that is, given an word $w \in L_2$ the transducer $A : L_2 \mapsto F^*$ returns $A(w) \in F^*$ where $A(w)$ is the adapted loanword in $\mathcal{F}$. We can employ formalisms used in formal language theory to perform the task.

Taking inspiration from O-COCOSDA paper and [3], we can use context-sensitive rewrite rules defined as an attribute grammar (AG) to implement the LA algorithm.

Since LA requires constraints from F and the orthography and phonetics of the w , we can embed these constraints as attributes required to be satisfied to perform the rewrite.

Hence, we will have a context-sensitive grammar (CSG) G , which defines the rewrite-rules as productions of the form

$$\text{rule_i}(\omega_i) \rightarrow \omega_j \quad | \quad \omega_i \models C$$

Informally, we will rewrite ω_i to ω_j if and only if ω_i satisfies each constraint in C .

III. PROPOSED ALGORITHM

IV. EXPERIMENTAL SETUP

V. RESULTS AND DISCUSSION

VI. CONCLUSION AND FUTURE WORK

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